Native-Token-Count Name-Channel Noninterference with Authenticated Role Review

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Abstract

Structural name-channel isolation can make model scores invariant to explicitly designated identity fields, but a segmentation-independent scalarwise codec can inflate transformer sequence length. We derive and implement a canonical-evidence factorization that uses exactly the original tokenizer's token count for retained evidence. The complete original named source is tokenized and embedded first; only then is one evidence sequence constructed and retokenized inside the model backend. This changes the computational boundary, not the underlying language model weights. We also add source-exact span validation, registered-reference coverage, explicit retention of same-spelled skills or third parties and two credential endorsements of the same resolved payload. On one frozen Qwen2.5-1.5B runtime, 108 controlled forwards yielded identical scores and 3,360 directly equal counterfactual public-stage arrays. All six native-reference hidden and vocabulary comparisons matched. Across 32 generated text cases, transformer tokens fell from 25,201 to 8,457, matching native evidence BPE exactly. A local 21-pair benchmark measured median service latencies of 334.4 ms versus 81.5 ms, with median paired speedup 4.10x. All 108 registered-omission controls and 36 review-conflict controls were rejected. Unknown aliases remain a deliberate negative control: authenticated agreement is not a proof of semantic role correctness. The work is a reproducible preprint draft, not universal automatic name discovery, whole-model unlearning or a racial-fairness deployment certificate.

1. Scope and motivation

The requested property is name noninterference: changing an applicant identity must not change a public score when job-relevant retained evidence stays fixed. Accurate grading is a separate property. A score can be invariant and wrong; a constant score is trivially invariant but provides no qualification sensitivity. We require the demonstration to remain nonconstant across evidence bodies, without claiming those scores establish actual hiring usefulness.

The prior local prototype embedded all original source fields, then excluded identity activations before attention. To avoid tokenization changes when public text was split across annotation nodes, it encoded each Unicode scalar separately. That convention was structurally convenient but expensive: the three reference bodies used roughly 4.5 times ordinary evidence BPE length. The present extension asks whether strong segmentation independence actually requires that cost.

Our answer is no, if the architecture is allowed to construct a canonical evidence sequence internally before its scoring transformer. This is not autonomous learned name erasure. It is a new explicit information-flow boundary with typed role metadata. The construction is related in purpose to earlier internal source/evidence routing in the same project; we do not claim literature- wide novelty for canonicalization or universal learned unlearning.

2. Related work and distinctions

Subword BPE [1] merges useful character/byte sequences and helps handle an open vocabulary. The standard tokenization pipeline distinguishes normalization, pre-tokenization, the subword model and post-processing [2]. Applying such a tokenizer independently to arbitrary pieces generally does not reproduce tokenizing their concatenation. This explains why deleting name-token rows from a context-dependent full-source BPE sequence does not always produce native evidence token IDs or positions.

Counterfactual fairness [3] motivates controlled changes to a sensitive factor, but our name-channel intervention is narrower than a causal model of race. Information-flow systems distinguish integrity of labels from confidentiality of outputs [4]. Our review signatures address later tampering, while the architectural theorem addresses dependence on designated identity inputs. Neither addresses every racial proxy or fairness of the underlying job policy.

Agreement coefficients [5] describe consistency among annotations, not their semantic correctness. Two signatures on an identical payload are authenticated agreement by two credentials; they are not evidence of two independent human judgments. We explicitly preserve the possibility of a shared missing alias. This extension evaluates mechanical annotation safeguards with a synthetic role oracle, not an invented human-review accuracy study.

3. Formal problem and previous encoding cost

Let D be a bounded source-exact list of identity/evidence text nodes. Let X(D) concatenate all original text, and Q(D) concatenate evidence text only. Let T be the fixed original deterministic tokenizer. The former scalarwise codec was:

```
C_scalar(Q) = T(c1) || ... || T(ck),
m(Q) = sum_i |T(ci)|,        n(Q) = |T(Q)|.
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If a codec must satisfy C(a+b)=C(a)‖C(b) for every possible scalar split, induction forces this scalar factorization. Whole-string BPE is generally not such a homomorphism; therefore retaining arbitrary independent-piece encoding while also retaining every native cross-boundary merge is an incompatible requirement in general. The overhead varies by script and text, not a universal constant. It should be measured, not inferred from English examples alone.

Three options are distinct. Fixed-field BPE is inexpensive but makes public field boundaries part of the equivalence contract. Scalarwise encoding permits arbitrary splits but loses merges. Canonicalizing retained evidence before one tokenizer call permits both arbitrary evidence segmentation and native evidence BPE. We implement the third option while preserving actual original-source embedding before canonicalization.

4. Internal canonical-evidence construction

The actual forward order is:

```
R = Emb(T(X(D)))                  # all original names embedded first
E = Q(D)                         # constructed inside model backend
H0 = Emb(T(E))                   # native evidence token embeddings
H_(l+1) = F_l(H_l; theta)        # original 28 blocks, no raw-source cache
S(D) = fixed_original_readout(H_L).
```

Raw R is evaluated before E is constructed. No anonymous text is supplied in place of the original source at the external encoding step. Nevertheless, ordinary code inside the model explicitly constructs E from reviewed roles. This fact must not be omitted when describing the method. The system has not learned to identify and ignore every raw name without supporting metadata.

The raw branch supplies no tensor, source length, cached state or position to the scoring branch. Original native evidence masks and positions start from T(E). The model uses its original embeddings, transformer, normalization and tied vocabulary head with the same FP32-input readout convention in all native comparisons. No learned adapter is installed in this extension.

5. Conditional proof and efficiency derivation

For accepted documents D,D' with Q(D)=Q(D'), deterministic tokenization yields T(Q(D))=T(Q(D')). Original evidence embeddings are equal. Each fixed transformer layer receives the same preceding tensor, positions and mask; induction gives equal hidden states at every scoring layer and an equal deterministic score. The proof covers accepted new identity values and arbitrary resegmentation of the retained evidence. It requires correct role assignments that preserve Q; it makes no accuracy assumption and no claim about unannotated name discovery.

The scoring transformer processes exactly n(Q) positions. This is equality with the original tokenizer on the same evidence, not a claim that greedy BPE is globally token-minimal over every possible vocabulary segmentation. The original source is also embedded, so total embedding visits are n(X)+n(Q). Review checks and two tokenizer calls also cost CPU work. A name-bearing raw CV can have different boundary merges from Q; we do not assert n(X)>=n(Q) universally.

A rough prefill scaling with L layers, width d and MLP width f is:

```
Cost(n) = O(L*(n*d^2 + n*d*f + n^2*d)).
```

Changing m to n reduces token-linear block terms by m/n and the attention term by (m/n)^2, while adding raw embedding and canonicalization overhead. These ratios are analytical component comparisons, not promised end-to-end speedups. The local latency benchmark measures the complete supported service calls.

The symbolic factorization specification asks for unequal outputs under equal evidence with arbitrary deterministic tokenizer/embedding/transformer/readout functions; Z3 returns UNSAT. A separate independent-piece token-count counterexample returns SAT. These small specifications check congruence and non-homomorphism assumptions, not program correctness or GPU semantics.

6. Stronger annotation integrity and omission checks

Reviewers annotate exact original source spans, preserving every input character. The signed payload contains a policy version, ordered roles, source hash, registered identity strings, explicit same-spelled retained meanings and a resolved status. Exact text/offsets, overlap, registry coverage and two known distinct credential signatures on the same payload are checked before model execution. Duplicate JSON keys, wrong signatures, changed policies, stale spans, unresolved proposals and conflicting payloads reject.

Registry detection supports casefolding, accents, full-width forms, combining characters, zero-width formatting characters and punctuation-separated names. Detection does not normalize source text used for scoring. Candidate spans remain unresolved until semantic review. A name matching the programming language Python or a third-party client requires a source-bound retained role and reason rather than automatic removal.

This mechanism checks declared textual coverage, not every nickname, indirect reference, transliteration or encoded identifier. Two credentials can be owned by one operator or share the same semantic mistake. They need distinct keys, but the program cannot prove two independent human judgments. The synthetic study explicitly measures none of those human properties.

Let A mean that roles are complete/correct and the retained evidence is fixed under the identity intervention; let B mean the structural runtime is correct. For an output violation V, the conditional theorem gives V impossible on A and B. Under a specified input-distribution probability model:

```
P(V) <= P(A^c union B^c) <= P(A^c) + P(B^c).
```

Neither probability is estimated from real CVs here. Two reviewer miss rates p1,p2 imply joint miss probability p1*p2 only with a justified independence assumption. Correlated joint misses can be as high as min(p1,p2). We do not claim that requiring two signatures squares the semantic error rate.

Character-level agreement, kappa and identity-character F1 are available as diagnostics. Kappa is undefined when its expected-agreement denominator is zero; the code returns null. High character agreement on a long CV can still conceal a short omitted name. Future evaluation needs independent adjudicated source truth and document-level omission/false-removal measurements [5].

7. Experimental design

The source-hashed plan is frozen before inference. Conformance crosses three generated evidence bodies, 12 identity strings and three source/segmentation variants: 108 full forwards. Names include several scripts, full-width letters, zero-width/NUL characters, a skill-name collision, an instruction-like string and a 1,000-character value. Every original source is embedded intact. Three forwards establish per-body references; 105 counterfactual forwards compare 32 public stages each: canonical IDs, evidence embeddings, 28 block outputs, final normalization and full final-position vocabulary logits. Direct array comparisons, rather than hash equality alone, check 3,360 counterfactual stages. Three native reference forwards separately compare final hidden states and full vocabulary arrays, giving six native-reference comparisons.

The token-count corpus contains 32 generated cases: eight language/style templates with four repetition lengths. It is not real applicant data or a population sample. Native evidence BPE, old scalar public tokens, raw-source BPE and total embedding-token visits are all recorded. No inputs are truncated. One old/scalar example exceeds its public-token limit; all canonical cases fit.

Latency uses two warm-ups per method/body and seven paired repetitions across three bodies (21 pairs, 42 timed calls), with shuffled order. Signing and model load are excluded; review verification, encoding, embedding, inference and per-call graph checks are included. The new service performs two signatures and registry checks; the old scalar service uses its earlier single-signature API. This is a service-package comparison, not a token-count-only ablation. Twelve additional warm-up calls are separate from the 108 conformance forwards.

Annotation evaluation uses 36 source-exact generator-oracle documents. Each has three identity nodes; each omission is tested separately (108 controls). Another 36 tests combine endorsements of different payloads. Two randomly generated machine credentials are used, not two human reviewers. An unregistered alias is retained as a deliberately unsolved negative control.

8. Results

All 108 conformance forwards preserved exact matched-name/resegmentation scores and public activations. All six native-evidence reference comparisons matched. The scorer remained nonconstant across the three evidence bodies; correctness of those scores is neither required nor claimed.

Measurement	Result
Scoring-transformer ratio to native evidence BPE	1.000x
Reference body 1: scalar -> canonical tokens	281 -> 62
Reference body 2: scalar -> canonical tokens	291 -> 63
Reference body 3: scalar -> canonical tokens	284 -> 64

Measurement	Result
Generated-corpus scalar -> canonical total	25,201 -> 8,457
Aggregate corpus transformer-token reduction	66.44%
Matched-name/segmentation score gap	0
Direct counterfactual array comparisons	3,360, all equal
Native evidence hidden and full-vocabulary checks	6, all equal
Omitted registered references blocked	108/108
Conflicting reviewed payload pairs blocked	36/36
CPU tests, including 2,000 Unicode property cases	52 pass

Across generated cases, old-scalar/native evidence ratios range from 1.39x to 6.37x, with mean per-case ratio 3.57x. The aggregate reduction above is computed from total token counts, not a mean of these ratios. The three John Miller reference raw-plus-canonical embedding counts are 133/135/137, versus old scalar raw embeddings 326/336/329. Extra source embedding work is not hidden by the native transformer count.

Median measured service time was 334.4 ms scalar versus 81.5 ms canonical. The median paired speedup was 4.10x, with interquartile range 4.01x-4.37x. These are local warm timings on one Apple Silicon runtime; no independent cross-device speedup or population confidence interval is asserted. The final factory smoke test passed the dual-review public API, process-wide lock, poisoned-service reuse rejection and final parameter/file integrity checks.

9. Limitations and negative controls

The strongest unresolved issue is correct semantic role annotation. In the synthetic study it is exact by construction. In arbitrary real text a name can denote an applicant, a client, a tool, an organization or an ambiguous referent. Shared mistaken endorsements do not remove that ambiguity. The explicit negative control preserves the alias Leaf when only John was registered; the program does not claim that this omission was automatically discovered.

Canonicalization is supporting code inside the model backend. It has the same role-based information-exclusion purpose as anonymization, while retaining the original named source embedding before the exclusion. It is not an autonomous model that has learned universal semantic name neutrality. Raw embeddings and stored name knowledge remain, and actual race is not identified or erased. Other CV proxies, population correlations and job-policy bias remain.

The proof compares accepted requests under a benign trusted runtime. Error, admission, timing, resource use and private audit observations are not protected. Canonical overflow occurs inside the backend and poisons that service; rebuild before reuse. The exact pinned checkpoint intentionally rejects other exports. Native-evidence computational parity includes native mistakes and does not establish useful hiring accuracy or native named-CV score preservation.

The public corpus and all role labels are generated, not independently annotated real applicants. The evidence bodies are few and repeated. No model training, independent replication, externally preregistered human-review study or new theory claim is supplied. The original project used public résumé/job datasets, but those records are not used or redistributed in this extension.

10. Reproducibility and publication status

The release includes a source-hashed plan, actual token/timing records, full synthetic score metadata, annotation controls, tests, symbolic specifications, read-only verifiers, the standalone PDF and Markdown source, an artifact manifest and CPU CI. Large activation arrays are compared during GPU execution but are not all retained; the CPU verifier replays records/coverage, not independent GPU inference. The tested model is a separate unadapted MLX affine-four-bit derivative of Qwen2.5-1.5B-Instruct, with exact parameter digest and file hashes in the package. Original base files and loaded parameters remain unchanged. Clef and other training jobs are not used or interrupted. Keys are random in memory and not saved.

This is a portfolio-ready research artifact and preprint draft, not an accepted or peer-reviewed publication, a DOI-bearing submission, a production hiring system or proof of academic novelty. Author review, independent replication, real annotation-quality studies and a venue-appropriate submission are still needed. Code, experiments and prose were developed with AI coding assistance; this is disclosed for review. The authored code/documentation are MIT-licensed; external model, tokenizer, dependency and paper terms remain with their owners.

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